

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

MACHINE LEARNING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
 - (3) Illustrate your answers with neat sketches.
 - (4) Use of Non – programmable calculator is permitted.
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1.
 - (a) A blood test for a Dengue has a sensitivity of 99% and a specificity of 97%. If 0.1% of the population has the condition, what is the probability that a person who tests positive actually has the condition ? 3(CO2)
 - (b) The parameters for multivariate Gaussian distribution classes are given as follows : $\mu_1 = [1, 0]$, $\Sigma_1 = [1, 0.2; 0.2, 1]$, $P(C_1) = 0.3$; $\mu_2 = [-1, 1]$, $\Sigma_2 = [1, 0; 0, 2]$, $P(C_2) = 0.3$; $\mu_3 = [0, 0]$, $\Sigma_3 = [2, 0; 0, 2]$, $P(C_3) = 0.4$. Given a new data point $x = [0, 1]$, determine the class to which this data point most likely belongs. 7(CO1,2)
 2.
 - (a) Write a Python script to read a CSV file named "Delhi.csv" and convert the read data into a DataFrame. Print the data types of each column and visualize the pairwise relationships in the dataset using a pair plot. Print the unique values in each categorical column and plot the count of each unique value using bar plots. Also unskew the data if skewness is greater than 1. 4(CO3,4)
 - (b) Write a Python script to read a CSV file named "student_grades.csv" and convert the read data into a DataFrame. Identify and encode categorical features using label encoding. Visualize the features using a pairplot. Apply t-SNE to reduce the dimensionality for visualization purposes. 6(CO3,4)

3. (a) Determine the optimal least risk decision boundaries for a feature x that is normally distributed for class C with mean=0 and std. dev.=2 and normally distributed for class D with mean=1 and std. dev.=1. The loss for misclassifying a sample from class C is INR 1000 and the loss for class D is INR 150. If $P(C)=0.5$ and $P(D)=0.5$, what is the decision boundary that minimizes the expected loss ? 6(CO2,4)
- (b) Write a python script to predict the next day rainfall in Delhi using "rainfall_Delhi.csv". Apply any two techniques with cross validation and print the error metrics. 4(CO3,4)
4. (a) Apply K-NN technique for $k=2$ using euclidean distance to classify a sample with a feature vector (1, 1) if samples from class A are at (3, 0), (4, 1) and (3, 2) and samples from class B are at (1, -1) and (1, -1.5). 4(CO2,4)
- (b) Write a Python script to split the data present in X (features) and Y (targets) into 70% train and 30% test set. Use decision trees and random forest trees with L1 and L2 regularization. Include the statement to plot the confusion matrix and print the Precision and Recall values. 6(CO3,4)
5. (a) Implement a K-means clustering algorithm using python and evaluate its effectiveness on a digits dataset. Use dimensionality reduction techniques (e.g., PCA) to visualize the clusters and support your evaluation. 7(CO2,4)
- (b) Explain the Gradient Descent algorithm and its variants with an example. 3(CO2)
6. (a) A healthcare provider is developing a machine learning system to assist in diagnosing diseases based on patient data. Describe the necessary components for the system with a block diagram. What features will you choose and how many will you use ? What are the challenges in this application and how will you address them ? 7(CO5)
- (b) Consider a scenario where autonomous vehicles are deployed in urban environments. Discuss whether the benefits of autonomous vehicles in terms of reducing traffic accidents and improving transportation efficiency outweigh the potential risks related to safety, liability and job displacement. Provide a justification for your stance. 3(CO1,2)

